Imports

```
# PyTorch
import torch
print(f"Is CUDA available? {torch.cuda.is_available()}")
print(f"GPU Name: {torch.cuda.get_device_name(0)}")
print(f"CUDA Version: {torch.version.cuda}")
import torchvision
import torchvision.transforms as transforms
import torch.nn as nn
import torch.nn.functional as nnFunc
import torch.amp as amp
# General
import numpy as np
import matplotlib.pyplot as plt
# Optimizer
import torch.optim as optim
# pretty print
from pprint import pprint
```

```
Is CUDA available? True
GPU Name: NVIDIA GeForce RTX 5050 Laptop GPU
CUDA Version: 13.0
```

Dataset

This code uses the CIFAR-10 dataset. (https://www.cs.toronto.edu/~kriz/cifar.html)

1. Transform setting

The original images are in PIL (0 to 255 for each pixel). First we convert it to a tensor format (0 to 1). Then to make the data to be zero centering, we make the range to be between -1 to 1. If we use the following equation for each data, we can make it zero centered: \begin{equation} output = \frac{input - mean}{std} \end{equation} This is being done by setting the `mean` and `standard deviation` to be 0.5.

```
transform = transforms.Compose([  transforms.RandomHorizontalFlip(),
                                  transforms.RandomCrop(32, padding=4),
                                  transforms.ToTensor(),
                                  transforms.Normalize((0.4914, 0.4822,
0.4465), (0.2023, 0.1994, 0.2010))
                                  ])
```

2. Making the training set

Now we load the actual dataset and make the training dataset.

1. Load CIFAR10 dataset (Download and prepare dataset)
    a. Put it in `./data` directory.
    b. Download it
    c. Do the specified transformation on the dataset
2. Create the data loader (pipelining settings)
    a. Brake it to batches with size of 32
    b. Shuffle the deck in every epoch
    c. Use 4 number of workers to make the training faster

```
batchSize = 512
numberOfWorkers = 0

trainSet = torchvision.datasets.CIFAR10(root='./data_CIFAR10',
                                        train=True,
                                        download=True,
                                        transform = transform)

trainLoader = torch.utils.data.DataLoader(trainSet,
                                          batch_size=batchSize,
                                          shuffle=True,
                                          num_workers=numberOfWorkers,
                                          pin_memory=True,
                                          persistent_workers=False)
```

3. Making the test set

Now making the test set. The procedure is the same as making the training set with some minor modifications.

```
testSet = torchvision.datasets.CIFAR10(root='./data_CIFAR10',
                                       train=False,
                                       download=True,
                                       transform=transform)

testLoader = torch.utils.data.DataLoader(testSet,
                                         batch_size=batchSize,
                                         shuffle=False,
                                         num_workers=numberOfWorkers,
                                         pin_memory=True,
                                         persistent_workers=False) #
Updated here too
viz_batchSize = 4
vizLoader = torch.utils.data.DataLoader(testSet,
batch_size=viz_batchSize, shuffle=True)
```

4. The classes

```
classes = ('plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog',
'horse', 'ship', 'truck')
pprint(classes)
```

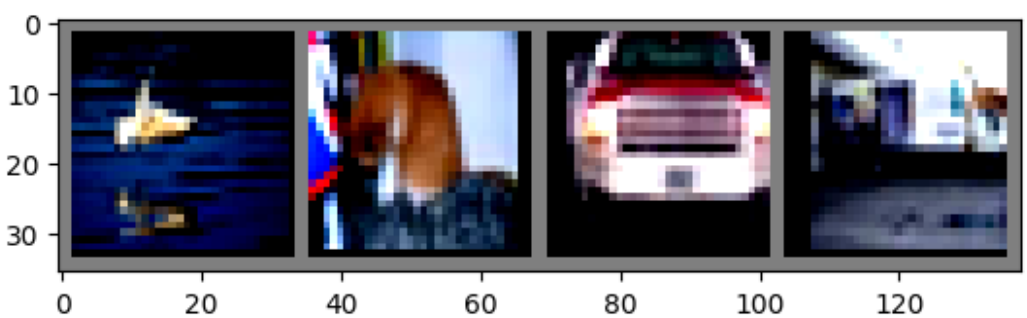
```
('plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse',
'ship', 'truck')
```

DEUBG1

Showing some of the images from the dataset.

```
# functions to show an image
def imshow(img):
  img = img / 2 + 0.5 # unnormalize
  npimg = img.numpy()
  plt.imshow(np.transpose(npimg, (1, 2, 0)))
  plt.show()
# get some random training images
dataiter = iter(vizLoader)
images, labels = next(dataiter)
# show images
imshow(torchvision.utils.make_grid(images))
# print labels
pprint(' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```

```
Clipping input data to the valid range for imshow with RGB data
([0..1] for floats or [0..255] for integers). Got range [-
0.71453285..1.8768656].
```




```
'bird  cat   truck truck'
```

5. Defining the CNN model

```
class Net(nn.Module):
  def __init__(self):
```

```
    super().__init__()
    self.conv1 = nn.Conv2d(3, 6, 5)
    self.pool1 = nn.AvgPool2d(2, 2)
    self.conv2 = nn.Conv2d(6, 16, 5)
    self.pool2 = nn.AvgPool2d(2, 2)
    self.fc1 = nn.Linear(16 * 5 * 5, 120)
    self.fc2 = nn.Linear(120, 84)
    self.fc3 = nn.Linear(84, 10)

  def forward(self, x):
    x = self.pool1(torch.tanh(self.conv1(x)))
    x = self.pool2(torch.tanh(self.conv2(x)))

    x = x.view(-1, 16 * 5 * 5)

    x = nnFunc.relu(self.fc1(x))
    x = nnFunc.relu(self.fc2(x))
    x = self.fc3(x)

    return x
```

6. Instansiating the CNN model

```
def init_weights(m):
    if isinstance(m, nn.Conv2d) or isinstance(m, nn.Linear):
        nn.init.kaiming_normal_(m.weight, mode='fan_out',
nonlinearity='relu')
        if m.bias is not None:
            nn.init.constant_(m.bias, 0)
net = Net()
net.apply(init_weights)
```

```
Net(
  (conv1): Conv2d(3, 6, kernel_size=(5, 5), stride=(1, 1))
  (pool1): AvgPool2d(kernel_size=2, stride=2, padding=0)
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (pool2): AvgPool2d(kernel_size=2, stride=2, padding=0)
  (fc1): Linear(in_features=400, out_features=120, bias=True)
  (fc2): Linear(in_features=120, out_features=84, bias=True)
  (fc3): Linear(in_features=84, out_features=10, bias=True)
)
```

7. Move the model to GPU

```
device = torch.device('cuda:0' if torch.cuda.is_available() else
'cpu')
print(device)
```

```
cuda:0
```

```
net.to(device)

Net(
  (conv1): Conv2d(3, 6, kernel_size=(5, 5), stride=(1, 1))
  (pool1): AvgPool2d(kernel_size=2, stride=2, padding=0)
  (conv2): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
  (pool2): AvgPool2d(kernel_size=2, stride=2, padding=0)
  (fc1): Linear(in_features=400, out_features=120, bias=True)
  (fc2): Linear(in_features=120, out_features=84, bias=True)
  (fc3): Linear(in_features=84, out_features=10, bias=True)
)
```

8. Degine loss function and optimizer

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(net.parameters(), lr=0.1, momentum=0.9)

scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer,
mode='max', factor=0.1, patience=10)
```

9. Traing the network

```
scaler = amp.GradScaler('cuda')

epochs = 100
printFrequency = len(trainLoader) // 4

for epoch in range(epochs):
    net.train() # Ensure model is in training mode
    running_loss = 0.0
    correct = 0
    total = 0

    # --- TRAINING PHASE ---
    for i, data in enumerate(trainLoader, 0):
        inputs, labels = data[0].to(device, non_blocking=True),
data[1].to(device, non_blocking=True)

        optimizer.zero_grad()

        with amp.autocast('cuda'):
            outputs = net(inputs)
            loss = criterion(outputs, labels)

        scaler.scale(loss).backward()
        scaler.unscale_(optimizer)
        torch.nn.utils.clip_grad_norm_(net.parameters(), max_norm=1.0)

        scaler.step(optimizer)
```

```
        scaler.update()

        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
        running_loss += loss.item()

        if i % printFrequency == (printFrequency - 1):
            accuracy = 100 * correct / total
            print(f'Epoch {epoch + 1} | Batch {i +
1}/{len(trainLoader)} | loss: {running_loss / printFrequency:.3f} |
acc: {accuracy:.2f}%')
            running_loss, correct, total = 0.0, 0, 0

    # --- VALIDATION PHASE ---
    net.eval()
    val_correct = 0
    val_total = 0
    with torch.no_grad(): # No gradients needed for validation
        for data in testLoader:
            images, labels = data[0].to(device), data[1].to(device)
            with amp.autocast('cuda'):
                outputs = net(images)
            _, predicted = torch.max(outputs.data, 1)
            val_total += labels.size(0)
            val_correct += (predicted == labels).sum().item()

    val_acc = 100 * val_correct / val_total
    current_lr = optimizer.param_groups[0]['lr']
    print(f'>> Epoch {epoch + 1} SUMMARY: Val Acc: {val_acc:.2f}% |
Current LR: {current_lr}')

    # --- AUTOMATIC LR DROP ---
    scheduler.step(val_acc)

print('Finished Training')
```

```
Epoch 1 | Batch 24/98 | loss: 3.063 | acc: 14.67%
Epoch 1 | Batch 48/98 | loss: 2.075 | acc: 25.06%
Epoch 1 | Batch 72/98 | loss: 1.974 | acc: 28.53%
Epoch 1 | Batch 96/98 | loss: 1.910 | acc: 30.32%
>> Epoch 1 SUMMARY: Val Acc: 32.30% | Current LR: 0.1
Epoch 2 | Batch 24/98 | loss: 1.851 | acc: 31.93%
Epoch 2 | Batch 48/98 | loss: 1.820 | acc: 33.81%
Epoch 2 | Batch 72/98 | loss: 1.751 | acc: 36.30%
Epoch 2 | Batch 96/98 | loss: 1.737 | acc: 36.17%
>> Epoch 2 SUMMARY: Val Acc: 37.54% | Current LR: 0.1
Epoch 3 | Batch 24/98 | loss: 1.689 | acc: 38.19%
Epoch 3 | Batch 48/98 | loss: 1.642 | acc: 39.54%
Epoch 3 | Batch 72/98 | loss: 1.664 | acc: 39.36%
```

```
Epoch 3 | Batch 96/98 | loss: 1.612 | acc: 41.13%
>> Epoch 3 SUMMARY: Val Acc: 41.57% | Current LR: 0.1
Epoch 4 | Batch 24/98 | loss: 1.598 | acc: 41.54%
Epoch 4 | Batch 48/98 | loss: 1.575 | acc: 42.54%
Epoch 4 | Batch 72/98 | loss: 1.545 | acc: 43.45%
Epoch 4 | Batch 96/98 | loss: 1.533 | acc: 44.03%
>> Epoch 4 SUMMARY: Val Acc: 44.83% | Current LR: 0.1
Epoch 5 | Batch 24/98 | loss: 1.513 | acc: 44.54%
Epoch 5 | Batch 48/98 | loss: 1.523 | acc: 44.38%
Epoch 5 | Batch 72/98 | loss: 1.526 | acc: 44.82%
Epoch 5 | Batch 96/98 | loss: 1.477 | acc: 45.74%
>> Epoch 5 SUMMARY: Val Acc: 45.07% | Current LR: 0.1
Epoch 6 | Batch 24/98 | loss: 1.489 | acc: 45.43%
Epoch 6 | Batch 48/98 | loss: 1.472 | acc: 45.79%
Epoch 6 | Batch 72/98 | loss: 1.474 | acc: 46.33%
Epoch 6 | Batch 96/98 | loss: 1.444 | acc: 47.72%
>> Epoch 6 SUMMARY: Val Acc: 48.70% | Current LR: 0.1
Epoch 7 | Batch 24/98 | loss: 1.446 | acc: 47.61%
Epoch 7 | Batch 48/98 | loss: 1.440 | acc: 47.80%
Epoch 7 | Batch 72/98 | loss: 1.460 | acc: 47.32%
Epoch 7 | Batch 96/98 | loss: 1.439 | acc: 48.06%
>> Epoch 7 SUMMARY: Val Acc: 48.49% | Current LR: 0.1
Epoch 8 | Batch 24/98 | loss: 1.427 | acc: 48.43%
Epoch 8 | Batch 48/98 | loss: 1.400 | acc: 49.38%
Epoch 8 | Batch 72/98 | loss: 1.410 | acc: 49.26%
Epoch 8 | Batch 96/98 | loss: 1.398 | acc: 49.27%
>> Epoch 8 SUMMARY: Val Acc: 49.56% | Current LR: 0.1
Epoch 9 | Batch 24/98 | loss: 1.398 | acc: 49.15%
Epoch 9 | Batch 48/98 | loss: 1.398 | acc: 49.80%
Epoch 9 | Batch 72/98 | loss: 1.382 | acc: 49.89%
Epoch 9 | Batch 96/98 | loss: 1.370 | acc: 50.48%
>> Epoch 9 SUMMARY: Val Acc: 48.69% | Current LR: 0.1
Epoch 10 | Batch 24/98 | loss: 1.360 | acc: 50.90%
Epoch 10 | Batch 48/98 | loss: 1.376 | acc: 50.76%
Epoch 10 | Batch 72/98 | loss: 1.351 | acc: 51.20%
Epoch 10 | Batch 96/98 | loss: 1.359 | acc: 50.89%
>> Epoch 10 SUMMARY: Val Acc: 50.60% | Current LR: 0.1
Epoch 11 | Batch 24/98 | loss: 1.354 | acc: 51.33%
Epoch 11 | Batch 48/98 | loss: 1.351 | acc: 51.11%
Epoch 11 | Batch 72/98 | loss: 1.339 | acc: 52.29%
Epoch 11 | Batch 96/98 | loss: 1.353 | acc: 51.01%
>> Epoch 11 SUMMARY: Val Acc: 50.87% | Current LR: 0.1
Epoch 12 | Batch 24/98 | loss: 1.334 | acc: 52.20%
Epoch 12 | Batch 48/98 | loss: 1.323 | acc: 52.68%
Epoch 12 | Batch 72/98 | loss: 1.339 | acc: 51.77%
Epoch 12 | Batch 96/98 | loss: 1.336 | acc: 51.87%
>> Epoch 12 SUMMARY: Val Acc: 51.24% | Current LR: 0.1
Epoch 13 | Batch 24/98 | loss: 1.322 | acc: 52.03%
Epoch 13 | Batch 48/98 | loss: 1.313 | acc: 52.80%
```

```
Epoch 13 | Batch 72/98 | loss: 1.315 | acc: 53.41%
Epoch 13 | Batch 96/98 | loss: 1.341 | acc: 51.34%
>> Epoch 13 SUMMARY: Val Acc: 51.87% | Current LR: 0.1
Epoch 14 | Batch 24/98 | loss: 1.324 | acc: 52.69%
Epoch 14 | Batch 48/98 | loss: 1.299 | acc: 52.87%
Epoch 14 | Batch 72/98 | loss: 1.279 | acc: 54.10%
Epoch 14 | Batch 96/98 | loss: 1.340 | acc: 52.01%
>> Epoch 14 SUMMARY: Val Acc: 52.97% | Current LR: 0.1
Epoch 15 | Batch 24/98 | loss: 1.283 | acc: 53.60%
Epoch 15 | Batch 48/98 | loss: 1.299 | acc: 53.80%
Epoch 15 | Batch 72/98 | loss: 1.289 | acc: 53.67%
Epoch 15 | Batch 96/98 | loss: 1.324 | acc: 52.57%
>> Epoch 15 SUMMARY: Val Acc: 53.27% | Current LR: 0.1
Epoch 16 | Batch 24/98 | loss: 1.297 | acc: 53.06%
Epoch 16 | Batch 48/98 | loss: 1.292 | acc: 53.67%
Epoch 16 | Batch 72/98 | loss: 1.283 | acc: 53.99%
Epoch 16 | Batch 96/98 | loss: 1.255 | acc: 55.26%
>> Epoch 16 SUMMARY: Val Acc: 55.12% | Current LR: 0.1
Epoch 17 | Batch 24/98 | loss: 1.279 | acc: 53.88%
Epoch 17 | Batch 48/98 | loss: 1.295 | acc: 53.85%
Epoch 17 | Batch 72/98 | loss: 1.276 | acc: 54.04%
Epoch 17 | Batch 96/98 | loss: 1.242 | acc: 55.55%
>> Epoch 17 SUMMARY: Val Acc: 54.42% | Current LR: 0.1
Epoch 18 | Batch 24/98 | loss: 1.239 | acc: 55.09%
Epoch 18 | Batch 48/98 | loss: 1.257 | acc: 55.24%
Epoch 18 | Batch 72/98 | loss: 1.268 | acc: 54.60%
Epoch 18 | Batch 96/98 | loss: 1.254 | acc: 55.35%
>> Epoch 18 SUMMARY: Val Acc: 54.78% | Current LR: 0.1
Epoch 19 | Batch 24/98 | loss: 1.257 | acc: 55.02%
Epoch 19 | Batch 48/98 | loss: 1.246 | acc: 56.27%
Epoch 19 | Batch 72/98 | loss: 1.247 | acc: 55.27%
Epoch 19 | Batch 96/98 | loss: 1.229 | acc: 55.95%
>> Epoch 19 SUMMARY: Val Acc: 52.88% | Current LR: 0.1
Epoch 20 | Batch 24/98 | loss: 1.236 | acc: 55.64%
Epoch 20 | Batch 48/98 | loss: 1.239 | acc: 55.67%
Epoch 20 | Batch 72/98 | loss: 1.229 | acc: 56.33%
Epoch 20 | Batch 96/98 | loss: 1.230 | acc: 55.60%
>> Epoch 20 SUMMARY: Val Acc: 53.72% | Current LR: 0.1
Epoch 21 | Batch 24/98 | loss: 1.217 | acc: 56.53%
Epoch 21 | Batch 48/98 | loss: 1.226 | acc: 56.27%
Epoch 21 | Batch 72/98 | loss: 1.216 | acc: 56.45%
Epoch 21 | Batch 96/98 | loss: 1.248 | acc: 55.12%
>> Epoch 21 SUMMARY: Val Acc: 55.20% | Current LR: 0.1
Epoch 22 | Batch 24/98 | loss: 1.220 | acc: 56.77%
Epoch 22 | Batch 48/98 | loss: 1.225 | acc: 55.73%
Epoch 22 | Batch 72/98 | loss: 1.194 | acc: 57.28%
Epoch 22 | Batch 96/98 | loss: 1.216 | acc: 56.85%
>> Epoch 22 SUMMARY: Val Acc: 52.99% | Current LR: 0.1
Epoch 23 | Batch 24/98 | loss: 1.235 | acc: 55.76%
```

```
Epoch 23 | Batch 48/98 | loss: 1.205 | acc: 56.71%
Epoch 23 | Batch 72/98 | loss: 1.199 | acc: 57.67%
Epoch 23 | Batch 96/98 | loss: 1.235 | acc: 55.88%
>> Epoch 23 SUMMARY: Val Acc: 56.77% | Current LR: 0.1
Epoch 24 | Batch 24/98 | loss: 1.220 | acc: 56.72%
Epoch 24 | Batch 48/98 | loss: 1.196 | acc: 57.14%
Epoch 24 | Batch 72/98 | loss: 1.210 | acc: 57.22%
Epoch 24 | Batch 96/98 | loss: 1.212 | acc: 56.70%
>> Epoch 24 SUMMARY: Val Acc: 55.89% | Current LR: 0.1
Epoch 25 | Batch 24/98 | loss: 1.202 | acc: 57.09%
Epoch 25 | Batch 48/98 | loss: 1.187 | acc: 58.02%
Epoch 25 | Batch 72/98 | loss: 1.195 | acc: 57.13%
Epoch 25 | Batch 96/98 | loss: 1.203 | acc: 57.38%
>> Epoch 25 SUMMARY: Val Acc: 56.24% | Current LR: 0.1
Epoch 26 | Batch 24/98 | loss: 1.201 | acc: 57.76%
Epoch 26 | Batch 48/98 | loss: 1.203 | acc: 56.32%
Epoch 26 | Batch 72/98 | loss: 1.203 | acc: 56.90%
Epoch 26 | Batch 96/98 | loss: 1.199 | acc: 57.36%
>> Epoch 26 SUMMARY: Val Acc: 55.61% | Current LR: 0.1
Epoch 27 | Batch 24/98 | loss: 1.167 | acc: 58.30%
Epoch 27 | Batch 48/98 | loss: 1.183 | acc: 57.63%
Epoch 27 | Batch 72/98 | loss: 1.191 | acc: 57.77%
Epoch 27 | Batch 96/98 | loss: 1.182 | acc: 57.78%
>> Epoch 27 SUMMARY: Val Acc: 57.15% | Current LR: 0.1
Epoch 28 | Batch 24/98 | loss: 1.191 | acc: 57.02%
Epoch 28 | Batch 48/98 | loss: 1.190 | acc: 57.76%
Epoch 28 | Batch 72/98 | loss: 1.168 | acc: 58.66%
Epoch 28 | Batch 96/98 | loss: 1.163 | acc: 58.81%
>> Epoch 28 SUMMARY: Val Acc: 57.82% | Current LR: 0.1
Epoch 29 | Batch 24/98 | loss: 1.160 | acc: 58.55%
Epoch 29 | Batch 48/98 | loss: 1.168 | acc: 57.58%
Epoch 29 | Batch 72/98 | loss: 1.160 | acc: 58.79%
Epoch 29 | Batch 96/98 | loss: 1.183 | acc: 57.78%
>> Epoch 29 SUMMARY: Val Acc: 56.54% | Current LR: 0.1
Epoch 30 | Batch 24/98 | loss: 1.161 | acc: 58.72%
Epoch 30 | Batch 48/98 | loss: 1.174 | acc: 58.23%
Epoch 30 | Batch 72/98 | loss: 1.185 | acc: 58.13%
Epoch 30 | Batch 96/98 | loss: 1.155 | acc: 58.68%
>> Epoch 30 SUMMARY: Val Acc: 57.74% | Current LR: 0.1
Epoch 31 | Batch 24/98 | loss: 1.156 | acc: 59.27%
Epoch 31 | Batch 48/98 | loss: 1.181 | acc: 58.18%
Epoch 31 | Batch 72/98 | loss: 1.157 | acc: 58.70%
Epoch 31 | Batch 96/98 | loss: 1.158 | acc: 58.32%
>> Epoch 31 SUMMARY: Val Acc: 57.47% | Current LR: 0.1
Epoch 32 | Batch 24/98 | loss: 1.143 | acc: 59.42%
Epoch 32 | Batch 48/98 | loss: 1.144 | acc: 58.99%
Epoch 32 | Batch 72/98 | loss: 1.147 | acc: 58.97%
Epoch 32 | Batch 96/98 | loss: 1.137 | acc: 59.40%
>> Epoch 32 SUMMARY: Val Acc: 57.86% | Current LR: 0.1
```

```
Epoch 33 | Batch 24/98 | loss: 1.147 | acc: 58.93%
Epoch 33 | Batch 48/98 | loss: 1.162 | acc: 58.83%
Epoch 33 | Batch 72/98 | loss: 1.147 | acc: 59.06%
Epoch 33 | Batch 96/98 | loss: 1.167 | acc: 57.95%
>> Epoch 33 SUMMARY: Val Acc: 57.76% | Current LR: 0.1
Epoch 34 | Batch 24/98 | loss: 1.149 | acc: 58.83%
Epoch 34 | Batch 48/98 | loss: 1.162 | acc: 58.78%
Epoch 34 | Batch 72/98 | loss: 1.155 | acc: 58.50%
Epoch 34 | Batch 96/98 | loss: 1.152 | acc: 58.68%
>> Epoch 34 SUMMARY: Val Acc: 56.87% | Current LR: 0.1
Epoch 35 | Batch 24/98 | loss: 1.150 | acc: 58.83%
Epoch 35 | Batch 48/98 | loss: 1.160 | acc: 58.86%
Epoch 35 | Batch 72/98 | loss: 1.134 | acc: 59.67%
Epoch 35 | Batch 96/98 | loss: 1.134 | acc: 59.73%
>> Epoch 35 SUMMARY: Val Acc: 57.10% | Current LR: 0.1
Epoch 36 | Batch 24/98 | loss: 1.139 | acc: 60.21%
Epoch 36 | Batch 48/98 | loss: 1.145 | acc: 58.76%
Epoch 36 | Batch 72/98 | loss: 1.137 | acc: 59.29%
Epoch 36 | Batch 96/98 | loss: 1.115 | acc: 60.92%
>> Epoch 36 SUMMARY: Val Acc: 59.46% | Current LR: 0.1
Epoch 37 | Batch 24/98 | loss: 1.143 | acc: 59.55%
Epoch 37 | Batch 48/98 | loss: 1.129 | acc: 59.37%
Epoch 37 | Batch 72/98 | loss: 1.153 | acc: 59.31%
Epoch 37 | Batch 96/98 | loss: 1.115 | acc: 60.21%
>> Epoch 37 SUMMARY: Val Acc: 57.51% | Current LR: 0.1
Epoch 38 | Batch 24/98 | loss: 1.125 | acc: 60.27%
Epoch 38 | Batch 48/98 | loss: 1.125 | acc: 59.42%
Epoch 38 | Batch 72/98 | loss: 1.141 | acc: 59.81%
Epoch 38 | Batch 96/98 | loss: 1.126 | acc: 59.78%
>> Epoch 38 SUMMARY: Val Acc: 59.43% | Current LR: 0.1
Epoch 39 | Batch 24/98 | loss: 1.112 | acc: 60.37%
Epoch 39 | Batch 48/98 | loss: 1.104 | acc: 61.04%
Epoch 39 | Batch 72/98 | loss: 1.117 | acc: 60.70%
Epoch 39 | Batch 96/98 | loss: 1.128 | acc: 59.38%
>> Epoch 39 SUMMARY: Val Acc: 59.50% | Current LR: 0.1
Epoch 40 | Batch 24/98 | loss: 1.113 | acc: 60.82%
Epoch 40 | Batch 48/98 | loss: 1.118 | acc: 60.57%
Epoch 40 | Batch 72/98 | loss: 1.117 | acc: 59.99%
Epoch 40 | Batch 96/98 | loss: 1.107 | acc: 60.82%
>> Epoch 40 SUMMARY: Val Acc: 59.00% | Current LR: 0.1
Epoch 41 | Batch 24/98 | loss: 1.113 | acc: 60.72%
Epoch 41 | Batch 48/98 | loss: 1.123 | acc: 59.79%
Epoch 41 | Batch 72/98 | loss: 1.105 | acc: 60.57%
Epoch 41 | Batch 96/98 | loss: 1.118 | acc: 60.33%
>> Epoch 41 SUMMARY: Val Acc: 59.34% | Current LR: 0.1
Epoch 42 | Batch 24/98 | loss: 1.125 | acc: 60.03%
Epoch 42 | Batch 48/98 | loss: 1.129 | acc: 59.71%
Epoch 42 | Batch 72/98 | loss: 1.114 | acc: 60.55%
Epoch 42 | Batch 96/98 | loss: 1.135 | acc: 59.69%
```

```
>> Epoch 42 SUMMARY: Val Acc: 57.72% | Current LR: 0.1
Epoch 43 | Batch 24/98 | loss: 1.147 | acc: 59.30%
Epoch 43 | Batch 48/98 | loss: 1.112 | acc: 60.74%
Epoch 43 | Batch 72/98 | loss: 1.121 | acc: 60.00%
Epoch 43 | Batch 96/98 | loss: 1.099 | acc: 61.18%
>> Epoch 43 SUMMARY: Val Acc: 59.93% | Current LR: 0.1
Epoch 44 | Batch 24/98 | loss: 1.093 | acc: 61.30%
Epoch 44 | Batch 48/98 | loss: 1.113 | acc: 60.58%
Epoch 44 | Batch 72/98 | loss: 1.118 | acc: 60.07%
Epoch 44 | Batch 96/98 | loss: 1.106 | acc: 60.07%
>> Epoch 44 SUMMARY: Val Acc: 59.65% | Current LR: 0.1
Epoch 45 | Batch 24/98 | loss: 1.070 | acc: 61.91%
Epoch 45 | Batch 48/98 | loss: 1.105 | acc: 60.83%
Epoch 45 | Batch 72/98 | loss: 1.095 | acc: 61.78%
Epoch 45 | Batch 96/98 | loss: 1.113 | acc: 60.47%
>> Epoch 45 SUMMARY: Val Acc: 59.99% | Current LR: 0.1
Epoch 46 | Batch 24/98 | loss: 1.104 | acc: 60.79%
Epoch 46 | Batch 48/98 | loss: 1.091 | acc: 61.58%
Epoch 46 | Batch 72/98 | loss: 1.116 | acc: 60.41%
Epoch 46 | Batch 96/98 | loss: 1.115 | acc: 60.29%
>> Epoch 46 SUMMARY: Val Acc: 58.88% | Current LR: 0.1
Epoch 47 | Batch 24/98 | loss: 1.106 | acc: 60.70%
Epoch 47 | Batch 48/98 | loss: 1.119 | acc: 60.51%
Epoch 47 | Batch 72/98 | loss: 1.103 | acc: 60.95%
Epoch 47 | Batch 96/98 | loss: 1.106 | acc: 60.90%
>> Epoch 47 SUMMARY: Val Acc: 58.99% | Current LR: 0.1
Epoch 48 | Batch 24/98 | loss: 1.074 | acc: 61.66%
Epoch 48 | Batch 48/98 | loss: 1.110 | acc: 60.45%
Epoch 48 | Batch 72/98 | loss: 1.076 | acc: 61.78%
Epoch 48 | Batch 96/98 | loss: 1.090 | acc: 61.80%
>> Epoch 48 SUMMARY: Val Acc: 58.54% | Current LR: 0.1
Epoch 49 | Batch 24/98 | loss: 1.090 | acc: 61.35%
Epoch 49 | Batch 48/98 | loss: 1.092 | acc: 60.99%
Epoch 49 | Batch 72/98 | loss: 1.093 | acc: 61.20%
Epoch 49 | Batch 96/98 | loss: 1.090 | acc: 61.22%
>> Epoch 49 SUMMARY: Val Acc: 60.80% | Current LR: 0.1
Epoch 50 | Batch 24/98 | loss: 1.068 | acc: 62.46%
Epoch 50 | Batch 48/98 | loss: 1.092 | acc: 61.06%
Epoch 50 | Batch 72/98 | loss: 1.063 | acc: 62.33%
Epoch 50 | Batch 96/98 | loss: 1.088 | acc: 61.69%
>> Epoch 50 SUMMARY: Val Acc: 59.80% | Current LR: 0.1
Epoch 51 | Batch 24/98 | loss: 1.092 | acc: 60.94%
Epoch 51 | Batch 48/98 | loss: 1.088 | acc: 61.60%
Epoch 51 | Batch 72/98 | loss: 1.049 | acc: 62.70%
Epoch 51 | Batch 96/98 | loss: 1.064 | acc: 62.35%
>> Epoch 51 SUMMARY: Val Acc: 60.63% | Current LR: 0.1
Epoch 52 | Batch 24/98 | loss: 1.080 | acc: 62.02%
Epoch 52 | Batch 48/98 | loss: 1.076 | acc: 61.78%
Epoch 52 | Batch 72/98 | loss: 1.083 | acc: 61.44%
```

```
Epoch 52 | Batch 96/98 | loss: 1.068 | acc: 62.37%
>> Epoch 52 SUMMARY: Val Acc: 59.96% | Current LR: 0.1
Epoch 53 | Batch 24/98 | loss: 1.086 | acc: 61.30%
Epoch 53 | Batch 48/98 | loss: 1.091 | acc: 61.51%
Epoch 53 | Batch 72/98 | loss: 1.098 | acc: 61.01%
Epoch 53 | Batch 96/98 | loss: 1.076 | acc: 61.77%
>> Epoch 53 SUMMARY: Val Acc: 59.07% | Current LR: 0.1
Epoch 54 | Batch 24/98 | loss: 1.080 | acc: 61.98%
Epoch 54 | Batch 48/98 | loss: 1.068 | acc: 62.06%
Epoch 54 | Batch 72/98 | loss: 1.090 | acc: 61.53%
Epoch 54 | Batch 96/98 | loss: 1.082 | acc: 61.65%
>> Epoch 54 SUMMARY: Val Acc: 61.35% | Current LR: 0.1
Epoch 55 | Batch 24/98 | loss: 1.075 | acc: 61.91%
Epoch 55 | Batch 48/98 | loss: 1.059 | acc: 62.38%
Epoch 55 | Batch 72/98 | loss: 1.096 | acc: 61.00%
Epoch 55 | Batch 96/98 | loss: 1.069 | acc: 62.04%
>> Epoch 55 SUMMARY: Val Acc: 61.20% | Current LR: 0.1
Epoch 56 | Batch 24/98 | loss: 1.057 | acc: 62.19%
Epoch 56 | Batch 48/98 | loss: 1.069 | acc: 62.48%
Epoch 56 | Batch 72/98 | loss: 1.080 | acc: 61.43%
Epoch 56 | Batch 96/98 | loss: 1.060 | acc: 62.51%
>> Epoch 56 SUMMARY: Val Acc: 59.50% | Current LR: 0.1
Epoch 57 | Batch 24/98 | loss: 1.074 | acc: 62.28%
Epoch 57 | Batch 48/98 | loss: 1.081 | acc: 61.62%
Epoch 57 | Batch 72/98 | loss: 1.072 | acc: 62.03%
Epoch 57 | Batch 96/98 | loss: 1.041 | acc: 62.88%
>> Epoch 57 SUMMARY: Val Acc: 60.99% | Current LR: 0.1
Epoch 58 | Batch 24/98 | loss: 1.072 | acc: 62.06%
Epoch 58 | Batch 48/98 | loss: 1.039 | acc: 63.63%
Epoch 58 | Batch 72/98 | loss: 1.064 | acc: 62.08%
Epoch 58 | Batch 96/98 | loss: 1.085 | acc: 61.17%
>> Epoch 58 SUMMARY: Val Acc: 60.18% | Current LR: 0.1
Epoch 59 | Batch 24/98 | loss: 1.078 | acc: 62.09%
Epoch 59 | Batch 48/98 | loss: 1.065 | acc: 61.93%
Epoch 59 | Batch 72/98 | loss: 1.068 | acc: 62.13%
Epoch 59 | Batch 96/98 | loss: 1.073 | acc: 62.09%
>> Epoch 59 SUMMARY: Val Acc: 60.74% | Current LR: 0.1
Epoch 60 | Batch 24/98 | loss: 1.039 | acc: 63.58%
Epoch 60 | Batch 48/98 | loss: 1.048 | acc: 62.72%
Epoch 60 | Batch 72/98 | loss: 1.086 | acc: 61.83%
Epoch 60 | Batch 96/98 | loss: 1.085 | acc: 61.58%
>> Epoch 60 SUMMARY: Val Acc: 60.42% | Current LR: 0.1
Epoch 61 | Batch 24/98 | loss: 1.048 | acc: 62.74%
Epoch 61 | Batch 48/98 | loss: 1.077 | acc: 61.68%
Epoch 61 | Batch 72/98 | loss: 1.067 | acc: 62.19%
Epoch 61 | Batch 96/98 | loss: 1.067 | acc: 62.66%
>> Epoch 61 SUMMARY: Val Acc: 61.09% | Current LR: 0.1
Epoch 62 | Batch 24/98 | loss: 1.022 | acc: 64.14%
Epoch 62 | Batch 48/98 | loss: 1.068 | acc: 62.21%
```

```
Epoch 62 | Batch 72/98 | loss: 1.060 | acc: 62.78%
Epoch 62 | Batch 96/98 | loss: 1.069 | acc: 62.89%
>> Epoch 62 SUMMARY: Val Acc: 59.90% | Current LR: 0.1
Epoch 63 | Batch 24/98 | loss: 1.040 | acc: 62.99%
Epoch 63 | Batch 48/98 | loss: 1.056 | acc: 62.65%
Epoch 63 | Batch 72/98 | loss: 1.048 | acc: 62.52%
Epoch 63 | Batch 96/98 | loss: 1.076 | acc: 61.89%
>> Epoch 63 SUMMARY: Val Acc: 60.30% | Current LR: 0.1
Epoch 64 | Batch 24/98 | loss: 1.044 | acc: 62.74%
Epoch 64 | Batch 48/98 | loss: 1.046 | acc: 62.86%
Epoch 64 | Batch 72/98 | loss: 1.069 | acc: 62.81%
Epoch 64 | Batch 96/98 | loss: 1.044 | acc: 63.15%
>> Epoch 64 SUMMARY: Val Acc: 61.31% | Current LR: 0.1
Epoch 65 | Batch 24/98 | loss: 1.032 | acc: 63.32%
Epoch 65 | Batch 48/98 | loss: 1.028 | acc: 63.92%
Epoch 65 | Batch 72/98 | loss: 1.055 | acc: 62.67%
Epoch 65 | Batch 96/98 | loss: 1.050 | acc: 62.50%
>> Epoch 65 SUMMARY: Val Acc: 61.68% | Current LR: 0.1
Epoch 66 | Batch 24/98 | loss: 1.046 | acc: 63.34%
Epoch 66 | Batch 48/98 | loss: 1.057 | acc: 63.07%
Epoch 66 | Batch 72/98 | loss: 1.040 | acc: 63.40%
Epoch 66 | Batch 96/98 | loss: 1.043 | acc: 63.44%
>> Epoch 66 SUMMARY: Val Acc: 61.80% | Current LR: 0.1
Epoch 67 | Batch 24/98 | loss: 1.035 | acc: 63.29%
Epoch 67 | Batch 48/98 | loss: 1.049 | acc: 62.97%
Epoch 67 | Batch 72/98 | loss: 1.060 | acc: 62.70%
Epoch 67 | Batch 96/98 | loss: 1.065 | acc: 62.59%
>> Epoch 67 SUMMARY: Val Acc: 60.37% | Current LR: 0.1
Epoch 68 | Batch 24/98 | loss: 1.054 | acc: 62.83%
Epoch 68 | Batch 48/98 | loss: 1.025 | acc: 63.96%
Epoch 68 | Batch 72/98 | loss: 1.052 | acc: 63.05%
Epoch 68 | Batch 96/98 | loss: 1.052 | acc: 62.32%
>> Epoch 68 SUMMARY: Val Acc: 61.30% | Current LR: 0.1
Epoch 69 | Batch 24/98 | loss: 1.022 | acc: 64.02%
Epoch 69 | Batch 48/98 | loss: 1.028 | acc: 63.35%
Epoch 69 | Batch 72/98 | loss: 1.049 | acc: 62.93%
Epoch 69 | Batch 96/98 | loss: 1.038 | acc: 63.21%
>> Epoch 69 SUMMARY: Val Acc: 61.97% | Current LR: 0.1
Epoch 70 | Batch 24/98 | loss: 1.027 | acc: 63.75%
Epoch 70 | Batch 48/98 | loss: 1.028 | acc: 63.94%
Epoch 70 | Batch 72/98 | loss: 1.052 | acc: 63.05%
Epoch 70 | Batch 96/98 | loss: 1.037 | acc: 62.95%
>> Epoch 70 SUMMARY: Val Acc: 62.23% | Current LR: 0.1
Epoch 71 | Batch 24/98 | loss: 1.015 | acc: 64.33%
Epoch 71 | Batch 48/98 | loss: 1.053 | acc: 62.93%
Epoch 71 | Batch 72/98 | loss: 1.026 | acc: 63.68%
Epoch 71 | Batch 96/98 | loss: 1.047 | acc: 63.44%
>> Epoch 71 SUMMARY: Val Acc: 60.53% | Current LR: 0.1
Epoch 72 | Batch 24/98 | loss: 1.045 | acc: 63.11%
```

```
Epoch 72 | Batch 48/98 | loss: 1.040 | acc: 63.05%
Epoch 72 | Batch 72/98 | loss: 1.033 | acc: 63.03%
Epoch 72 | Batch 96/98 | loss: 1.046 | acc: 62.55%
>> Epoch 72 SUMMARY: Val Acc: 61.29% | Current LR: 0.1
Epoch 73 | Batch 24/98 | loss: 1.035 | acc: 63.60%
Epoch 73 | Batch 48/98 | loss: 1.023 | acc: 63.75%
Epoch 73 | Batch 72/98 | loss: 1.024 | acc: 63.93%
Epoch 73 | Batch 96/98 | loss: 1.024 | acc: 63.80%
>> Epoch 73 SUMMARY: Val Acc: 62.62% | Current LR: 0.1
Epoch 74 | Batch 24/98 | loss: 1.030 | acc: 63.81%
Epoch 74 | Batch 48/98 | loss: 1.046 | acc: 63.00%
Epoch 74 | Batch 72/98 | loss: 1.043 | acc: 63.37%
Epoch 74 | Batch 96/98 | loss: 1.040 | acc: 63.48%
>> Epoch 74 SUMMARY: Val Acc: 62.07% | Current LR: 0.1
Epoch 75 | Batch 24/98 | loss: 1.023 | acc: 64.25%
Epoch 75 | Batch 48/98 | loss: 1.038 | acc: 63.31%
Epoch 75 | Batch 72/98 | loss: 1.026 | acc: 64.02%
Epoch 75 | Batch 96/98 | loss: 1.017 | acc: 63.75%
>> Epoch 75 SUMMARY: Val Acc: 61.91% | Current LR: 0.1
Epoch 76 | Batch 24/98 | loss: 1.034 | acc: 62.90%
Epoch 76 | Batch 48/98 | loss: 1.032 | acc: 63.65%
Epoch 76 | Batch 72/98 | loss: 1.034 | acc: 63.76%
Epoch 76 | Batch 96/98 | loss: 1.007 | acc: 64.26%
>> Epoch 76 SUMMARY: Val Acc: 61.70% | Current LR: 0.1
Epoch 77 | Batch 24/98 | loss: 1.011 | acc: 64.13%
Epoch 77 | Batch 48/98 | loss: 1.037 | acc: 63.61%
Epoch 77 | Batch 72/98 | loss: 1.042 | acc: 63.12%
Epoch 77 | Batch 96/98 | loss: 1.033 | acc: 63.64%
>> Epoch 77 SUMMARY: Val Acc: 62.41% | Current LR: 0.1
Epoch 78 | Batch 24/98 | loss: 1.005 | acc: 64.39%
Epoch 78 | Batch 48/98 | loss: 1.026 | acc: 63.92%
Epoch 78 | Batch 72/98 | loss: 1.015 | acc: 64.40%
Epoch 78 | Batch 96/98 | loss: 1.020 | acc: 63.95%
>> Epoch 78 SUMMARY: Val Acc: 61.81% | Current LR: 0.1
Epoch 79 | Batch 24/98 | loss: 1.033 | acc: 63.32%
Epoch 79 | Batch 48/98 | loss: 1.009 | acc: 64.44%
Epoch 79 | Batch 72/98 | loss: 1.027 | acc: 63.88%
Epoch 79 | Batch 96/98 | loss: 1.029 | acc: 63.80%
>> Epoch 79 SUMMARY: Val Acc: 61.50% | Current LR: 0.1
Epoch 80 | Batch 24/98 | loss: 1.025 | acc: 63.23%
Epoch 80 | Batch 48/98 | loss: 1.023 | acc: 64.31%
Epoch 80 | Batch 72/98 | loss: 1.028 | acc: 64.09%
Epoch 80 | Batch 96/98 | loss: 1.007 | acc: 64.91%
>> Epoch 80 SUMMARY: Val Acc: 62.67% | Current LR: 0.1
Epoch 81 | Batch 24/98 | loss: 1.008 | acc: 64.18%
Epoch 81 | Batch 48/98 | loss: 1.013 | acc: 64.26%
Epoch 81 | Batch 72/98 | loss: 1.023 | acc: 64.44%
Epoch 81 | Batch 96/98 | loss: 1.030 | acc: 63.68%
>> Epoch 81 SUMMARY: Val Acc: 62.59% | Current LR: 0.1
```

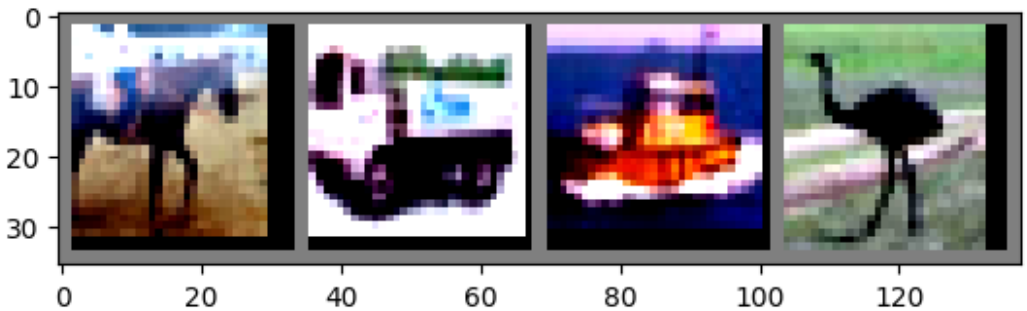
```
Epoch 82 | Batch 24/98 | loss: 1.008 | acc: 64.71%
Epoch 82 | Batch 48/98 | loss: 1.029 | acc: 63.49%
Epoch 82 | Batch 72/98 | loss: 1.020 | acc: 63.93%
Epoch 82 | Batch 96/98 | loss: 1.011 | acc: 64.18%
>> Epoch 82 SUMMARY: Val Acc: 61.55% | Current LR: 0.1
Epoch 83 | Batch 24/98 | loss: 1.026 | acc: 63.53%
Epoch 83 | Batch 48/98 | loss: 1.026 | acc: 63.92%
Epoch 83 | Batch 72/98 | loss: 1.017 | acc: 63.91%
Epoch 83 | Batch 96/98 | loss: 1.008 | acc: 64.62%
>> Epoch 83 SUMMARY: Val Acc: 61.56% | Current LR: 0.1
Epoch 84 | Batch 24/98 | loss: 1.014 | acc: 63.79%
Epoch 84 | Batch 48/98 | loss: 1.001 | acc: 64.66%
Epoch 84 | Batch 72/98 | loss: 1.008 | acc: 64.41%
Epoch 84 | Batch 96/98 | loss: 1.022 | acc: 63.91%
>> Epoch 84 SUMMARY: Val Acc: 62.88% | Current LR: 0.1
Epoch 85 | Batch 24/98 | loss: 0.996 | acc: 64.58%
Epoch 85 | Batch 48/98 | loss: 1.018 | acc: 64.15%
Epoch 85 | Batch 72/98 | loss: 1.002 | acc: 64.49%
Epoch 85 | Batch 96/98 | loss: 1.025 | acc: 63.72%
>> Epoch 85 SUMMARY: Val Acc: 62.18% | Current LR: 0.1
Epoch 86 | Batch 24/98 | loss: 1.007 | acc: 64.51%
Epoch 86 | Batch 48/98 | loss: 1.042 | acc: 62.90%
Epoch 86 | Batch 72/98 | loss: 1.029 | acc: 63.65%
Epoch 86 | Batch 96/98 | loss: 1.003 | acc: 65.00%
>> Epoch 86 SUMMARY: Val Acc: 62.85% | Current LR: 0.1
Epoch 87 | Batch 24/98 | loss: 0.987 | acc: 64.88%
Epoch 87 | Batch 48/98 | loss: 1.034 | acc: 62.91%
Epoch 87 | Batch 72/98 | loss: 1.018 | acc: 64.24%
Epoch 87 | Batch 96/98 | loss: 0.987 | acc: 65.04%
>> Epoch 87 SUMMARY: Val Acc: 62.56% | Current LR: 0.1
Epoch 88 | Batch 24/98 | loss: 0.988 | acc: 65.18%
Epoch 88 | Batch 48/98 | loss: 1.013 | acc: 64.10%
Epoch 88 | Batch 72/98 | loss: 1.000 | acc: 64.19%
Epoch 88 | Batch 96/98 | loss: 1.007 | acc: 64.81%
>> Epoch 88 SUMMARY: Val Acc: 63.39% | Current LR: 0.1
Epoch 89 | Batch 24/98 | loss: 1.034 | acc: 63.90%
Epoch 89 | Batch 48/98 | loss: 0.995 | acc: 64.76%
Epoch 89 | Batch 72/98 | loss: 1.015 | acc: 64.22%
Epoch 89 | Batch 96/98 | loss: 1.015 | acc: 63.78%
>> Epoch 89 SUMMARY: Val Acc: 63.21% | Current LR: 0.1
Epoch 90 | Batch 24/98 | loss: 0.995 | acc: 64.53%
Epoch 90 | Batch 48/98 | loss: 0.992 | acc: 65.67%
Epoch 90 | Batch 72/98 | loss: 1.012 | acc: 64.24%
Epoch 90 | Batch 96/98 | loss: 1.010 | acc: 63.72%
>> Epoch 90 SUMMARY: Val Acc: 62.27% | Current LR: 0.1
Epoch 91 | Batch 24/98 | loss: 0.992 | acc: 65.34%
Epoch 91 | Batch 48/98 | loss: 0.998 | acc: 64.63%
Epoch 91 | Batch 72/98 | loss: 0.997 | acc: 65.44%
Epoch 91 | Batch 96/98 | loss: 1.000 | acc: 64.46%
```

```
>> Epoch 91 SUMMARY: Val Acc: 62.84% | Current LR: 0.1
Epoch 92 | Batch 24/98 | loss: 0.995 | acc: 64.79%
Epoch 92 | Batch 48/98 | loss: 1.011 | acc: 64.47%
Epoch 92 | Batch 72/98 | loss: 0.997 | acc: 64.87%
Epoch 92 | Batch 96/98 | loss: 1.001 | acc: 65.10%
>> Epoch 92 SUMMARY: Val Acc: 63.24% | Current LR: 0.1
Epoch 93 | Batch 24/98 | loss: 1.002 | acc: 64.47%
Epoch 93 | Batch 48/98 | loss: 0.996 | acc: 65.09%
Epoch 93 | Batch 72/98 | loss: 0.985 | acc: 65.23%
Epoch 93 | Batch 96/98 | loss: 1.005 | acc: 64.69%
>> Epoch 93 SUMMARY: Val Acc: 61.96% | Current LR: 0.1
Epoch 94 | Batch 24/98 | loss: 1.005 | acc: 65.00%
Epoch 94 | Batch 48/98 | loss: 0.996 | acc: 64.39%
Epoch 94 | Batch 72/98 | loss: 0.996 | acc: 64.40%
Epoch 94 | Batch 96/98 | loss: 1.016 | acc: 64.09%
>> Epoch 94 SUMMARY: Val Acc: 61.58% | Current LR: 0.1
Epoch 95 | Batch 24/98 | loss: 0.988 | acc: 65.25%
Epoch 95 | Batch 48/98 | loss: 0.974 | acc: 65.33%
Epoch 95 | Batch 72/98 | loss: 1.001 | acc: 64.26%
Epoch 95 | Batch 96/98 | loss: 0.997 | acc: 64.88%
>> Epoch 95 SUMMARY: Val Acc: 63.68% | Current LR: 0.1
Epoch 96 | Batch 24/98 | loss: 0.975 | acc: 65.70%
Epoch 96 | Batch 48/98 | loss: 0.994 | acc: 65.06%
Epoch 96 | Batch 72/98 | loss: 0.994 | acc: 65.19%
Epoch 96 | Batch 96/98 | loss: 0.995 | acc: 64.74%
>> Epoch 96 SUMMARY: Val Acc: 63.10% | Current LR: 0.1
Epoch 97 | Batch 24/98 | loss: 0.971 | acc: 65.70%
Epoch 97 | Batch 48/98 | loss: 1.014 | acc: 63.69%
Epoch 97 | Batch 72/98 | loss: 1.004 | acc: 64.54%
Epoch 97 | Batch 96/98 | loss: 1.004 | acc: 64.66%
>> Epoch 97 SUMMARY: Val Acc: 63.35% | Current LR: 0.1
Epoch 98 | Batch 24/98 | loss: 0.992 | acc: 65.23%
Epoch 98 | Batch 48/98 | loss: 0.996 | acc: 65.04%
Epoch 98 | Batch 72/98 | loss: 0.979 | acc: 65.13%
Epoch 98 | Batch 96/98 | loss: 0.996 | acc: 64.78%
>> Epoch 98 SUMMARY: Val Acc: 63.63% | Current LR: 0.1
Epoch 99 | Batch 24/98 | loss: 0.986 | acc: 65.72%
Epoch 99 | Batch 48/98 | loss: 0.975 | acc: 65.54%
Epoch 99 | Batch 72/98 | loss: 1.012 | acc: 63.92%
Epoch 99 | Batch 96/98 | loss: 1.006 | acc: 64.41%
>> Epoch 99 SUMMARY: Val Acc: 62.79% | Current LR: 0.1
Epoch 100 | Batch 24/98 | loss: 1.012 | acc: 64.31%
Epoch 100 | Batch 48/98 | loss: 1.011 | acc: 64.18%
Epoch 100 | Batch 72/98 | loss: 1.010 | acc: 64.29%
Epoch 100 | Batch 96/98 | loss: 1.014 | acc: 64.28%
>> Epoch 100 SUMMARY: Val Acc: 63.05% | Current LR: 0.1
Finished Training
```

10. Test the network on the test data

```
dataIter = iter(vizLoader)
images, labels = next(dataIter)
# print images
imshow(torchvision.utils.make_grid(images))
print('GroundTruth: ', ' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```

```
Clipping input data to the valid range for imshow with RGB data
([0..1] for floats or [0..255] for integers). Got range [-
0.71453285..1.8768656].
```




```
GroundTruth:  horse truck ship  bird
```

```
images, labels = images.to(device), labels.to(device)

outputs = net(images)
_, predicted = torch.max(outputs, 1)

print('Predicted: ', ' '.join(f'{classes[predicted[j].item()]:5s}' for
j in range(len(predicted))))
```

```
Predicted:  horse truck ship  bird
```

11. Test the network on the whole dataset

```
test_loss = 0.0
correct = 0
total = 0

with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)

        outputs = net(images)

        # 1. Calculate the loss for this batch
        loss = criterion(outputs, labels)
```

```
        test_loss += loss.item()

        # 2. Calculate accuracy as before
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# Calculate the average loss over all batches
avg_test_loss = test_loss / len(testLoader)

print(f'Test Loss: {avg_test_loss:.4f}')
print(f'Test Accuracy: {100 * correct / total:.2f}%')
```

```
Test Loss: 1.0545
Test Accuracy: 63.67%
```

12. Check performance within each class

```
# prepare to count predictions for each class
correct_pred = {classname: 0 for classname in classes}
total_pred = {classname: 0 for classname in classes}

# again no gradients needed
with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)
        outputs = net(images)
        _, predictions = torch.max(outputs, 1)
        # collect the correct predictions for each class
        for label, prediction in zip(labels, predictions):
            if label == prediction:
                correct_pred[classes[label]] += 1
            total_pred[classes[label]] += 1

# print accuracy for each class
for classname, correct_count in correct_pred.items():
    accuracy = 100 * float(correct_count) / total_pred[classname]
    print(f'Accuracy for class: {classname:5s} is {accuracy:.1f} %')
```

```
Accuracy for class: plane is 70.4 %
Accuracy for class: car   is 73.2 %
Accuracy for class: bird  is 43.0 %
Accuracy for class: cat   is 39.9 %
Accuracy for class: deer  is 53.3 %
Accuracy for class: dog   is 57.2 %
Accuracy for class: frog  is 73.7 %
Accuracy for class: horse is 69.4 %
Accuracy for class: ship  is 78.4 %
Accuracy for class: truck is 72.4 %
```